$$\lim_{x \to -3} \frac{x^3+27}{x+3} = \lim_{x \to -3} \frac{\cancel{(x+3)}(x^2+3x+9)}{\cancel{x+3}} = \lim_{x \to -3} (-3)^2 + 3(-3) + 9$$

$$x^3 + 3^3$$

$$(x+3(x^2+x3+3^2)$$

$$= \lim_{x \to -3} \cancel{9-9} + 9 = 9$$

$$\lim_{x \to 0} \frac{x^2 + 7x}{x} = \lim_{x \to 0} \frac{\cancel{x}(x+7)}{\cancel{x}} = \lim_{x \to 0} x + 7 = 7$$

$$\lim_{x \to 1} \frac{x^2-3x+2}{x^2-4x+3} = \lim_{x \to 1} \frac{\cancel{(x-1)}(x-2)}{\cancel{(x-1)}(x-3)} = \lim_{x \to 1} \frac{x-2}{x-3} = \lim_{x \to 1} \frac{1-2}{1-3} = \frac{-1}{-2} = \frac{1}{2}$$

x^2-3x+2

$x \quad -1$

$x \quad -2$

$(x-1)(x-2$

x^2-4x+3

$x \quad -1$

$x \quad -3$

$(x-1)(x-3)$

$$\lim_{x\to 4} \frac{\sqrt{2x+1}-3}{\sqrt{x-2}-\sqrt{2}} \cdot \frac{\sqrt{x-2}+\sqrt{2}}{\sqrt{x-2}+\sqrt{2}} \cdot \frac{\sqrt{2x+1}+3}{\sqrt{2x+1}+3} = \lim_{x\to 4} \frac{(2x+1-3^2)(\sqrt{x-2}+\sqrt{2})}{x-2-2(\sqrt{2x+1}+3)}$$

$$= \lim_{x\to 4} \frac{\overbrace{\cancel{(2x-8)}}^{x-4}(\sqrt{x-2}+\sqrt{2})}{\cancel{x-4}(\sqrt{2x+1}+3)} = \lim_{x\to 4} \frac{\overbrace{\cancel{(x-4)}}^{0}(\sqrt{x-2}+\sqrt{2})}{\sqrt{2x+1}+3}$$

$$= \lim_{x\to 4}$$

$$= \lim_{x\to 4} \underline{\underline{0}}$$